

Venuste NDIKUMANA

FULL-STACK WEB & MOBILE DEVELOPMENT

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PROFILE

Full-stack and mobile software engineering. Independently designs, builds, and ships complete production-style systems end-to-end from database schema and REST/real-time APIs to cross-platform Flutter clients while applying secure coding practices and integrating AI APIs into real projects. Combines hands-on LLM integration experience with strong written documentation, structured critical analysis, and a sustained domain interest in cybersecurity and AI governance in specialized contexts (e.g., correctional systems).

Attended the Ejo Labs Summer Talent Program 2026 to grow applied AI and product-engineering skills in a fast-moving, mission-driven team.

TECHNICAL SKILLS

Languages: Dart, JavaScript, TypeScript, Python, SQL

Mobile & Frontend: Flutter (BLoC architecture), React, Material 3, cross-platform UI

Backend & APIs: Node.js, Express.js, REST API design, Socket.IO, Django

Data & Cloud: PostgreSQL, SQLite, Firebase (Firestore, Admin SDK, Cloud Messaging)

AI Integration: OpenAI API (GPT-4o-mini) for real-time translation and applied NLP features

Security & Auth: JWT, bcrypt, rate limiting, input validation, Helmet.js, role-based access control

Practices & Tools: Git/GitHub, system design, Linux server deployment (Nginx, Unicorn), Tailwind CSS

AI TRAINING & EVALUATION SKILLS

- **Data Annotation & Evaluation:** Reviewed, validated, and corrected GPT-4o-mini translation outputs across English, French, Kinyarwanda, and Swahili while building CampChat, identifying mistranslations and edge cases.
- **Prompt Engineering:** Designed and iterated prompts driving GPT-4o-mini's on-demand and auto-translate features, tuning for accuracy and context-appropriate output.
- **Critical Thinking & Feedback Delivery:** Authored a 43-page written analysis and a 12-slide deck ("AI for Safer Prisons") assessing AI applications and risks in correctional contexts; delivered structured architectural feedback across the SORP and DGL system designs.
- **Written Communication:** Produced extensive technical documentation, including a 22-section academic report, system-design specs, and API/event contracts, aimed at both technical and non-technical readers.
- **Domain Expertise:** BSc focus in Computer Science and Information Security; sustained applied interest in cybersecurity and correctional-systems technology.
- **Remote Collaboration & Adaptability:** Independently scoped, built, and shipped four full end-to-end systems (CampChat, DGL, PCMS, SORP) without a team, adapting quickly to new domains and requirements.
- **Attention to Detail:** Applied defense-in-depth security practices (JWT auth, bcrypt hashing, rate limiting, parameterized queries, input validation) consistently across every project.

- **Ethical Awareness in AI:** Grounded AI-feature design in secure-by-default and privacy-conscious practices; explored ethical and safety considerations of AI deployment in sensitive institutional (correctional) settings.

WORK EXPERIENCE

Full-Stack Developer *Self-Directed / Independent*

2024 – Present

- Independently designed, built, and shipped multiple full-stack and mobile systems end-to-end spanning backend APIs, real-time systems, relational and NoSQL database design, and cross-platform Flutter clients (see Selected Engineering Projects below).
- Applied secure coding practices (JWT authentication, bcrypt, input validation, rate limiting) and integrated third-party AI APIs (OpenAI) into production-style applications, including prompt design and output review.

SELECTED FEATURED PROJECTS

CampChat: AI-Powered Multilingual Messaging Platform

- Designed and built a secure, cross-platform real-time messaging app end-to-end: a Flutter/BLoC mobile client backed by a Node.js/Express API, with Socket.IO powering live 1-on-1 and group messaging, delivery/read receipts, typing indicators, and presence.
- Integrated the OpenAI API (GPT-4o-mini) to deliver on-demand and auto-translate messaging across English, French, Kinyarwanda, and Swahili solving a real cross-language communication gap for local users, including prompt design and output evaluation.
- Implemented a defense-in-depth security layer: JWT access/refresh token authentication, bcrypt password hashing, per-route rate limiting, parameterized SQL queries, Helmet.js headers, and express-validator input checks.
- Built an offline-first data layer using SQLite (sqlite) on-device synced against a normalized PostgreSQL schema, plus media sharing (image, video, audio, document) and a Material 3 light/dark UI.

Driver Garage Linker (DGL): Driver Garage Marketplace Platform

- Architected a multi-module cross-platform workspace connecting a Flutter mobile client (separate driver and garage-owner experiences) to a Node.js/Express backend, using Socket.IO for real-time trip/status updates and the Firebase Admin SDK for backend services.
- Modeled the core domain (drivers, garages, trips, vehicles) and designed the account-verification and role-based access flow connecting the two user types.
- Set up Firestore indexing and hosting configuration, and defined the project's lint/test quality-check workflow for both the Flutter client and the Node API.

Personal Computer Maintenance System (PCMS): Full-Stack Service Management App

- Built a full-stack web application, split into independent backend and frontend modules, to manage computer repair and maintenance service workflows covering customer intake, technician assignment, and service tracking.
- Combined a JavaScript service layer with Python-based components, applying separation-of-concerns between the presentation layer and business logic.

SORP: Staff Operation & Reporting Platform (Final-Year Capstone, System Design)

- Authored a complete system design for an enterprise daily-duty and incident-reporting mobile platform: Firestore data schema, Flutter client architecture, Node.js API and Socket.IO event contracts, offline-first sync strategy, and a hierarchical role-based access control model. Repository setup pending.

EDUCATION

BSc. Computer Science and Information Security

University of Rwanda, College of Science and Technology · Focus: Full-Stack & Mobile Development

Secondary Education (A-Level): Combination: PCM (Physics, Chemistry, Mathematics)

Groupe Scolaire Sainte Bernadette de Save

Secondary Education (O-Level)

Groupe Scolaire Sainte Dominique Savio Kibangu

CERTIFICATIONS

- Advanced Networking Operations: Internet Society, 2026
- AI and Prompt Engineering: EjoLabs, 2026
- AI Career Essentials: ALX, September 2024
- Junior Cybersecurity: Cisco Networking Academy, August 2024
- Introduction to Cybersecurity: Cisco Networking Academy, May 2024

LANGUAGES

Kinyarwanda: Native · English: Intermediate · Swahili: Basic